Chapter 2

Microscopy

Chapter Outline

2.1 Historical Background

2.2 Principles of Microscopy

2.3 Bright Field Microscope

2.4 Dark Field Microscope

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Microorganisms are very small and cannot be viewed by human eye. The microscope helps in observing the microbial world which exists in a wide range of sizes. The prokaryotes (bacteria and archae) are smaller (~ 0.4-10µm) and the eukaryotes are larger (~ or >10µm). The word microscope is derived from the Latin word *micro*, which means small, and the Greek word skopos means *to look at.*

Learning Objectives

After studying this chapter the student will be able,

- *To know the properties of light and lens.*
- *To know the science of image formation in brightfield microscopy.*
- *To understand the design of light microscope.*
- *To learn and compare the principle, instrumentation and working of brightfield and darkfield microscopy.*

2.1 Historical Background

Antony Van Leeuwenhoek (1632-1723) was the first person to use a simple microscope with one lens similar to a magnifying glass. The lens is capable of 50X to 300X magnification.

Robert Hooke, built compound microscopes with multiple lenses. In 17th century, Dutch spectacle maker Zaccharias Janssen is given the credit for making first compound microscope. However, the early compound microscopes were poor in quality. In 1830, Joseph Jackson Lister (the father of Joseph Lister who practised antiseptic surgery) made significant development which resulted in the invention of modern compound microscope used in microbiology today.

2.2 Principles of Microscopy

All kind of microscopes use visible light to observe specimens. Light has a number of properties that affect our ability to visualise objects.

2.2.1 Properties of Light

Light is a part of the wide spectrum of electromagnetic radiation from the sun. It is a form of energy. The most important property of light is wavelength (the length